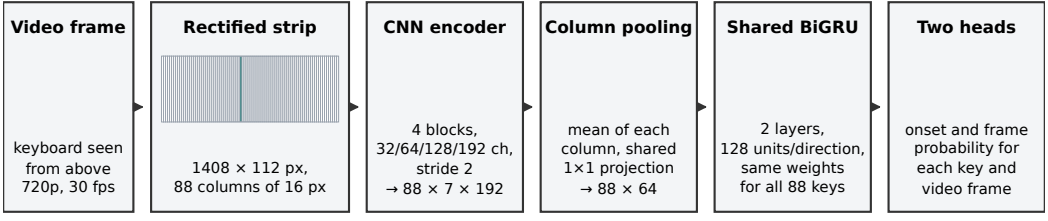
Video frame
keyboard seen from above 720p, 30 fps
Rectified strip
1408 × 112 px, 88 columns of 16 px
CNN encoder
4 blocks, 32/64/128/192 ch, stride 2 → 88 × 7 × 192
Column pooling
mean of each column, shared 1×1 projection → 88 × 64
Shared BiGRU
2 layers, 128 units/direction, same weights for all 88 keys
Two heads
onset and frame probability for each key and video frame